IMPORTANT INSTRUCTIONS :

- **Pattern of the Entrance Examination**:-
  Paper containing 180 objective type questions ,from Biology, Physics and Chemistry
- **Use Blue/Black Ball Point Pen only** to darken the appropriate circle. Answers marked with pencil would not be evaluated.
- Each item carries **4marks**. For each correct response the candidate will get 4 marks. For each incorrect response **1mark** will be deducted from the total score

1. The terms couplet and lead are associated with a taxonomic aid namely
   (1) Herbarium
   (2) Botanical keys
   (3) Museum
   (4) Manuals
2. Which one among the following is against the rules of ICBN ?
   (1) Scientific names are in the Latin and need to be italicized
   (2) When handwritten generic name and specific name should be separately underlined
   (3) The name of author is cited in abbreviated form and is not underlined
   (4) Both generic and specific names should be written starting with small letters
3. Grevy's zebra and plains zebra are native animals of savannahs. Both of them feed on grass species. These two competitors were niche differentiated and started to feed on 15 grass species. Grevy's zebra started to feed mainly on 10 species and plains zebra on 5 species. – This concept explains
   (1) Competitive exclusion
   (2) Competitive interference
   (3) Resource partitioning
   (4) Predation pressure
4. Which of the following hormones help to synthesize milk and eject milk respectively?
   (1) FSH and LH
   (2) Progesterone and estrogen
   (3) LH and LTH
   (4) Prolactin and oxytocin
5. Match column – I with column – II

Column – I	Column – II
a) Lichens	i) Acellular infectious proteins
b) Mycorrhiza	ii) Mutualistic association of fungi and vascular plant
c) Archaea	iii)Pioneers in xerosere
d) Prion	iv) Non pathogenic extremophiles

   (1) a – ii ; b – iii ; c – iv ; d – i
   (2) a – iii ; b – iv ; c – iii ; d – i
   (3) a – iii ; b – ii ; c – iv ; d – i
   (4) a – ii ; b – iv ; c – iii ; d – i

6. Match the items given in Column - I with those in Column - II and choose the correct option

Column-I	Column-II
(a) Rennin	(i) Enzyme activator
(b) Enterokinase	(ii) Infant enzyme
(c) Bile	(iii) Facilitated transport
(d) Glucose	(iv) Activates lipase

(1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
(2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(3) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
(4) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

7. Which of the following statements is NOT true regarding ascomycetes ?
(1) Conidia are asexual spores produced exogenously
(2) Yeast is unicellular fungi reproducing asexually by budding
(3) Rusts and smuts are caused by pathogenic fungi of this class
(4) *Claviceps* is a source of many alkaloids and LSD

8. Which of the animals are worm like but true coelomates?
(1) Platyhelminthes
(2) Aschelminthes
(3) Hemichordata
(4) Cyclostomata

9. Female gametophyte is retained within integumented megasporangium in
(1) Pteridophytes
(2) Bryophytes
(3) Spermatophytes
(4) Thallophytes

10. Match the following animals with respective salient feature

Column-I	Column-II
(a) Lamprey	(i) Comb plates for locomotion
(b) Snail	(ii) Statocysts, as balancing organs
(c) Prawn	(iii) Radula, as rasping organ
(d) Sea walnuts	(iv) Spawn in freshwater

(1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
(2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(3) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
(4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

11. Which one among the following shows alternate/spiral phyllotaxy ?
(1) Mustard
(2) *Nerium*
(3) *Calotropis*
(4) *Psidium*

12. Excitable tissue at rest is least permeable to
(1) Sodium
(2) Potassium
(3) Magnesium
(4) Chloride

13. Placentation in which ovules develop along the ventral suture of a monocarpellary ovary is
(1) Parietal placentation
(2) Free central placentation
(3) Basal placentation
(4) Marginal placentation

14. Bicarpellary ovary with false septum and unequal length of stamen in its flowers is seen in
    (1) *Solanum*
    (2) *Brassica*
    (3) *Aloe*
    (4) *Sesbania*
15. Which one among the following is incorrect?
    (1) Apical meristems are responsible for indeterminate growth in plants
    (2) Intercalary meristems help in regeneration of damaged growing grass stems
    (3) Lateral meristems help the plant to increase in its diameter
    (4) Vascular cambium in dicot stem arise from pericycle and conjunctive tissue by dedifferentiation
16. A man with blood group B, cannot be the biological father of child with an AB blood type if
    (1) The mother is blood type O
    (2) The mother is blood type A
    (3) The mother is blood type AB
    (4) The mother is Rh positive
17. Erythropoietin secretion occurs when
    (1) Decreased tissue pO_2 concentration
    (2) Increased tissue pH
    (3) Increased tissue pCO_2 concentration
    (4) Increased haemoglobin
18. Read the following statements
    (a) If the cones are stimulated equally, a sensation of white light is produced
    (b) When rods are not stimulated, they are found in depolarized state
    (c) Opsin is the protein found in both rods and cones
    (d) Iris is the visible coloured portion of that helps the formation of image on it

    In the above, the correct statements are
    (1) a, b and c
    (2) b, c and d
    (3) c, d
    (4) a only
19. Which one among the following in bark is NOT a derivative of redifferentiation ?
    (1) Bast
    (2) Cork
    (3) Phellogen
    (4) Phelloderm
20. Marasmus is a
    (1) Genetic disorder
    (2) Nutritional disorder
    (3) Contagious disease
    (4) Age related problem
21. Which is NOT associated with phloem in angiosperms ?
    (1) Tyloses
    (2) Companion cells
    (3) Sieve tubes
    (4) Fibers
22. Identify the *incorrect* combination from the following
    (1) Menstrual phase – Fall in estrogen and progesterone
    (2) Ovulatory phase – LH surge
    (3) Luteal phase – Rise in FSH
    (4) Follicular phase – Rise in estrogen

23. Which one of the statements given below is NOT true about annual rings in tree ?
(1) Prominent in temperate regions
(2) Contain spring and autumn wood
(3) Produced from phellogen by redifferentiation
(4) Formed due to differential activity of cambium

24. Zygomatic bone(s) is/are
(1) Facial paired bones
(2) Cranial unpaired bone
(3) Facial unpaired bone
(4) Cranial paired bones

25. Stomata in a grass leaf close during dry weather. Select the most appropriate reason from the following
(1) Flaccidity of bulliform cells
(2) Turgidity of dumbbell shaped guard cells
(3) Turgidity of bulliform cells
(4) Flaccidity of dumbbell shaped guard cells

26. Which of the following are *not* a set of homologous structures?
(1) Scales of onion and spines of *Opuntia*
(2) Wing of bird and forelimb of mammal
(3) Rhizome of ginger and Beetroot
(4) Mouthparts of mosquito and housefly

27. Pick the correct match from the following ?
(1) Passive transport – Energy driven
(2) Guttation – Water vapour lost through stoma
(3) Osmosis – Solvent movement into hypertonic solution
(4) Active transport – Non selective

28. Identify the *incorrect* statement
(1) The unpolluted water is characteristically rich in DO and low in BOD
(2) Higher the BOD, higher would be DO
(3) The polluted water has high values of BOD
(4) Water for drinking should have a BOD less than 1

29. *Nitrosomonas* is helpful in carrying out
(1) Denitrification
(2) Nitrification
(3) Nitrogen fixation
(4) Ammonification

30. Who reported that in green sulphur bacteria hydrogen from a suitable oxidisable compound reduce CO_2 to carbohydrates ?
(1) Engelmann
(2) Sachs
(3 Neil
(4) Hill

31. In Hatch and Slack pathway the primary CO_2 acceptor in bundle sheath cells is
(1) Phosphoenol pyruvic acid
(2) Oxalo acetic acid
(3) RuBP
(4) Phosphoglyceric acid

32. Frog shows similarity with cockroach but differs from earthworm in having
(1) Metamerism
(2) Complete gut
(3) Exoskeleton
(4) Dioecious nature

33. Identify the *mismatched* combination

Column-I	Column-II
(1) Alpha thalassemia	Chromosome-16
(2) Phenylketonuria	Chromosome-12
(3) Sickle cell anaemia	Chromosome-11
(4) Haemophilia	Chromosome-21

34. Select the correct statement regarding MOET
    (1) An elite cow is administered with estrogen and progesterone instead of FSH
    (2) Fertilized eggs are removed from the cow surgically
    (3) Genetic makeup of formed calves is similar to surrogate mother
    (4) Super ovulation results the production of 6-8 eggs per cycle

35. Where is ubiquinone located in a plant cell ?
    (1) Thylakoid membrane
    (2) Inner mitochondrial membrane
    (3) Mitochondrial matrix
    (4) Stroma of chloroplast

36. Application of 2,4,D is a very useful technique to make the weed free lawns. This is because
    (1) It enhances growth of apical bud
    (2) It prevents leaf drop at early stage
    (3) It kills broad leaved dicot weeds
    (4) It delays senescence in leaves

37. Presence of less number of birds in Greenland than to Columbia explains
    (1) Niche specialization is more in poles
    (2) Species diversity decreases from equator to poles
    (3) Competition is less in poles
    (4) No predators for birds in poles

38. Match the items given in Column - I with those in Column - II and choose the correct option

Column-I	Column-II
(a) Stabilizing selection	(i) More individuals acquire value other than mean character value
(b) Directional selection	(ii) More individuals acquire peripheral character value at both the ends
(c) Disruptive selection	(iii) Selective individuals whereby man intervenes in the breeding of species to produce desired traits
(d) Artificial selection	(iv) More individuals acquire mean character value

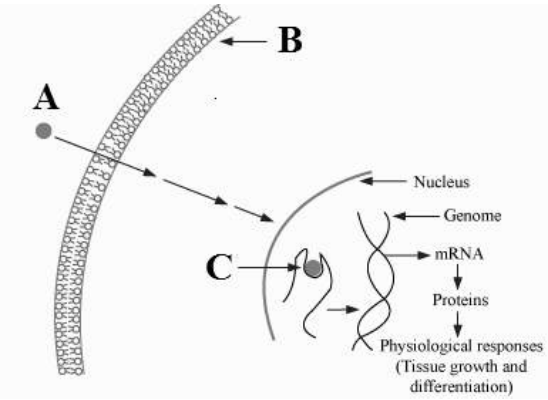
    (1) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)
    (2) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
    (3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
    (4) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

39. Which plant growth regulator is needed in order to increase the internodal length of vegetative shoot just before flowering ?
    (1) Gibberellins
    (2) Auxins
    (3) Cytokinins
    (4) ABA

40. A type of lymphocytes that found in innate immunity to constitute cellular barriers is
    (1) Natural killer cells
    (2) Killer cells
    (3) Polymorphonuclear leucocytes
    (4) Langerhans cells

41. When necessary inductive photoperiod is present hormonal substance responsible for flowering migrate from
(1) Shoot apices to leaves
(2) Leaves to shoot apices
(3) Shoot apices to flowers
(4) Leaves to root apices

42. Identify A, B and C in diagrammatic representation of the mechanism of hormone action



Select the correct from the following
(1) A-Estrogen; B-Ovarian wall; C-Cyclic AMP
(2) A-FSH; B-Ovarian wall; C-Hormone-receptor complex
(3) A-Estrogen; B-Uterine wall; C-Hormone-receptor complex
(4) A-Thyroxin; B-Thyroid gland vesicle; C-Hormone-receptor complex

43. Which one given below is <u>wrongly</u> paired ?
(1) Virchow – *Omnis cellula-e-cellula*
(2) Ramachandran – Triple helical structure of collagen
(3) Leeuwenhoek – Nucleus
(4) Schleiden & Schwann – Cell theory

44. Statement – I : Active synthesis of rRNA occur in non membranous nucleoplasmic structure namely nucleolus
Statement – II : Both DNA and RNA present inside mitochondria and chloroplast are single stranded and circular in shape
(1) Statement-I and Statement-II are true
(2) Statement-I is true but Statement-II is false.
(3) Statement-I is false but Statement-II is true.
(4) Statement-I and Statement-II are false

45. Identify the *mismatched* combination
(1) Sternum – Flat bone on the ventral midline of thorax
(2) Knee joint – Synovial joint between femur and tibio-fibula
(3) Ear ossicles – Malleus, incus and stapes found in middle ear
(4) Myasthenia gravis – An autoimmune disorder affecting usually cardiac muscle

46. All these glands release their components to outside the body except
(1) Cardiac glands of mammal
(2) Oil gland of bird
(3) Sebaceous glands of rabbit
(4) Accessory glands of earthworm

47. Cell organelle present in highest number in protein synthesizing cells and secretory cells respectively are
(1) Golgi complex and lysosomes
(2) Ribosomes and chloroplast
(3) Golgi complex and ribosomes
(4) Ribosomes and Golgi complex

48. Read the following and choose the correct combination

Column-I	Column-II	Column-III
a. Hypo thalamus	p. Fore brain	w. Inter connect different regions of brain and spinal cord
b. Limbic system	q. Mid brain	x. Emotional brain
c. Pons Varolii	r. Hind brain	y. Involuntary centres
d. Medulla oblongata		z. Thermo regulation

(1) a-p-w
(2) b-q-x
(3) c-r-z
(4) d-r-y

49. Which one among the following could be organic (or) inorganic in nature ?
(1) Metal ion
(2) Apoenzyme
(3) Cofactor
(4) All the above

50. Junctions that perform cementing to keep neighbouring cells together are
(1) Tight junctions
(2) Adhering junctions
(3) Gap junctions
(4) Synaptic junctions

51. Match column – I with column – II

Column – I	Column – II
a) GLUT – IV	i) Lectin
b) Concanavalin – A	ii) Phospholipid
c) Lecithin	iii) Cell cycle progression
d) Cyclin	iv) Carrier protein

(1) a – iv ; b – i ; c – iii ; d – ii
(2) a – iii ; b – i ; c – ii ; d – iv
(3) a – iv ; b – ii ; c – i ; d – iii
(4) a – iv ; b – i ; c – ii ; d – iii

52. Which of the following is a heritable disorder of X-chromosome?
(1) Alport syndrome
(2) Jacob syndrome
(3) Down syndrome
(4) SCID syndrome

53. What would be the number of bivalents in the meiocyte, if 30 chromosomes are present in each nucleus of a monosporic type of embyrosac ?
(1) 30
(2) 60
(3) 120
(4) 45

54. Oral Pills usually
(1) Inhibit ovulation and implantation
(2) Inhibit menstruation and implantation
(3) Inhibit sperms by the release of toxins
(4) Inhibit basal metabolism in woman

55. Pick the <u>wrong</u> match from the following
(1) S phase – Centriole duplication
(2) G_2 phase – Cell growth and protein synthesis
(3) G_o stage – Quiescent stage
(4) M phase – DNA replication

56. In family planning methods the female is sterilized by
(1) Oopherectomy
(2) Hysterectomy
(3) Vasectomy
(4) Tubectomy

57. Which one of the following statement regarding post fertilisation development in a flowering plant is incorrect ?
(1) Zygote → Embryo
(2) Ovary → Fruit
(3) Nucellus → Endosperm
(4) Ovules → Seeds

58. Variation in gene frequencies within populations can occur by the chance rather than by natural selection is termed
(1) Pancmictic mating
(2) Genetic migration
(3) Genetic mapping
(4) Genetic drift

59. Match column – I with column – II

Column – I	Column – II
a) Pseudocarp	i) Mango
b) Parthenocarp	ii) Apple
c) Eucarp	iii) Grass
d) Apomixis	iv) Banana

(1) a – iv ; b – ii ; c – i ; d – iii
(2) a – ii ; b – iv ; c – i ; d – iii
(3) a – iii ; b – i ; c – iv ; d – iii
(4) a – i ; b – iv ; c – iii ; d – ii

60. The following is an invertebrate with closed circulation
(1) *Peripatus*
(2) *Periplaneta*
(3) *Pheretima*
(4) *Pinctada*

61. What is the fate of male gametes discharged into synergid during double fertilisation in a flowering plant ?
(1) One fuse with egg and other degenerate
(2) One fuse with synergid and other with egg
(3) One fuse with egg and other with antipodals
(4) One fuse with egg and other with central cell nuclei

62. In *Antirrhinum* (Snapdragon) monohybrid cross is conducted between two parent plants with red and white flowers. All F_1 plants produce intermediate phenotype (pink flowers). Result of test cross progeny is
(1) Red and white flowered in 1 : 1 ratio
(2) All pink flowered
(3) Red, pink and white in 1 : 2 : 1 ratio
(4) Pink and white in 1 : 1 ratio

63. Phenotype represented by four different genotypes in F_2 generation of dihybrid cross is
(1) Round green
(2) Wrinkled green
(3) Round yellow
(4) Wrinkled yellow

64. The wall of blood vessels has ---A--- muscle fibers and they are ---B--- in their nature
(1) A-Spindle shaped; B-Voluntary
(2) A-Cylindrical; B-Involuntary
(3) A-Branched net like; B-Voluntary
(4) A-Fusiform; B-Involuntary

65. Orchids show bewildering diversity of floral patterns many of which have evolved to attract the right pollinators like
(1) Arboreal mammals
(2) Bumblebee birds
(3) Bumblebees
(4) White ants

66. Ejaculatory duct of male human receives the following
(1) Vas efferens and duct of seminal vesicle
(2) Vas deferens and duct of seminal vesicle
(3) Vas deferens and ducts of prostate gland
(4) Vas efferens and ducts of prostate gland

67. In *Drosophila*, recombinant phenotype percentage between body colour and eye colour determining genes is
(1) Zero
(2) 1.3 %
(3) 37.2 %
(4) 62.8 %

68. Experimental evidence for semiconservative type of DNA replication was given by
(1) Watson and Crick
(2) Messelson and Stahl
(3) Hershey and Chase
(4) Jacob and Monod

69. In human foetus the limbs and digits are developed by the end of
(1) 14 weeks
(2) First trimester
(3) 6th month
(4) 8 weeks

70. Identify the *mismatched* combination from the following
(1) Catalytic converter: CO and NO_X
(2) Scrubber: SO_2
(3) Landfills: Radioactive elements
(4) Electrostatic precipitator: PM

71. Pick the wrong match with regard to enzyme and the type of RNA transcribed from it ?
(1) RNA polymerase I – rRNA
(2) RNA polymerase II – snRNA
(3) RNA polymerase III – tRNA
(4) RNA polymerase (bacteria) – All three types of RNA's

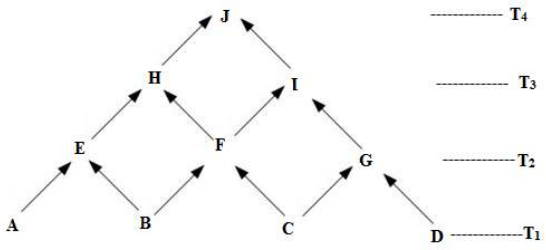
72. Different sequences of codons producing same sequence of amino acid in a protein is due to
(1) Universal nature of genetic code
(2) Specificity in genetic code
(3) Non overlapping nature of genetic code
(4) Degeneracy in genetic code

73. A physician advised to a woman to undergo 'mammogram' for screening, later he recommended to have a positron Emission Tomography scan, can you guess the doctor suspects what disease in her
(1) Breast cancer
(2) Colo - rectal block
(3) Ovarian cysts
(4) Cervical cancer

74. Functional residual capacity in the lungs of human is derived from 'p' and 'q'. The correct choices for 'p' and 'q' are
(1) p-RV and q-ERV
(2) p-TV and q-RV
(3) p-TV and q-IRV
(4) p-RV and q-IRV

75. Statement – I : Lactose regulate switching on and switching off of *lac* operon
Statement – II : '*i*' gene is a regulatory gene and it encode for a repressor protein both in the presence and absence of lactose in medium
(1) Statement-I and Statement-II are true
(2) Statement-I is true but Statement-II is false.
(3) Statement-I is false but Statement-II is true.
(4) Statement-I and Statement-II are false

76. What can be correct for following food web?



(1) I is climax consumer
(2) E is a primary consumer
(3) D is a detritivore
(4) H is a herbivore

77. Raw materials can be converted into specific products in large scale using microbial cells in
(1) Autoclave
(2) Bio reactor
(3) Sludge digester
(4) Laminal air flow chamber

78. Which of the following is always upright pyramid?
(1) Energy pyramid
(2) Number pyramid
(3) Biomass pyramid
(4) Age pyramid

79. Genes encoding antibiotic resistance used in selection of transformants in *r*DNA technology are called
(1) Recombinant proteins
(2) Molecular scissors
(3) Selectable markers
(4) Clones

80. The alkaline pH in the gut of the pest given below activate the protoxin to active *Bt* toxin encoded by the gene *cry I Ab* is
(1) Cotton boll worm
(2) Corn borer
(3) Dragon flies
(4) *Meloidogyne*

81. Sporozoites are the infective stages of
(1) *Ascaris lumbricoides*
(2) *Plasmodium vivax*
(3) *Entamoeba histoltyica*
(4) *Salmonella typhi*

82. Identify the correct statements
(a) Predator helps in maintaining species diversity in a community
(b) Decomposition is largely an anaerobic process
(c) Number of trophic levels in GFC is never restricted
(d) Some organisms of DFC are prey to GFC animals
(1) (a), (b) and (c)
(2) (a) and (d)
(3) (a), (b) and (d)
(4) (c) and (d)

83. Match column I with column II

Column – I	Column – II
a) Recognition sequence	i) Annealing sticky ends
b) Exonuclease	ii)Thermostable enzyme
c) Ligase	iii) Remove nucleotides from ends
d) *Taq* polymerase	iv) Palindrome for restriction enzyme activity

(1) a – iv ; b – iii ; c – ii ; d – i
(2) a – iv ; b – iii ; c – i ; d – ii
(3) a – iii ; b – ii ; c – iv ; d – i
(4) a – ii ; b – iv ; c – iii ; d – i

84. Ketone bodies are added to the urine of
(1) Cardiac patient
(2) Kidney patient
(3) Diabetic patient
(4) Healthy person with normal diet

85. Pick the wrong match
(1) *Trichoderma* – Biocontrol agent
(2) *Oscillatoria* – Biofertilizer
(3) *Aspergillus*– Aspartic acid
(4) Atlas 66 – Biofortification

86. Read the following with respect to human embryo
(a) Skull is close to skull of baby chimpanzee
(b) Avails oxygen by pulmonary breathing
(c) Collects glucose from mother by using its hPL
(d) Exhibits non-functional gills in its early stage
(e) Has fused coccyx as like in adult during the development of vertebrae

The correct features
(1) (a) and (b)
(2) (a), (c) and (d)
(3) (b), (c) and (e)
(4) (d) and (e)

87. Match column I with column II

Column – I	Column – II
a. Wheat	i. Pusa komal
b. Cauliflower	ii. Himgiri
c. Chilli	iii. Pusa shubhra
d. Cow pea	iv. Pusa sadabahar

(1) a – iv, b – iii, c – ii, d – i
(2) a – ii, b – iii, c – iv, d – i
(3) a – iii, b – iv, c – i, d – ii
(4) a – iv, b – ii, c – i, d – iii

88. Identify the *incorrect* statement
(1) Heroin is formed by acetylation of morphine
(2) Cocaine is extracted from *Erythroxylum* species
(3) Ganja interferes dopamine transmission
(4) Morphine is an effective sedative and pain killer

89. In ecological succession from pioneer to climax community, the biomass
(1) Decreases
(2) Increases and then decreases
(3) No relation
(4) Increases

90. Transgenic animals have been used
(1) For testing safety of vaccines
(2) For testing toxicity of drugs
(3) To produce useful biological products
(4) All the above

91. In an experiment the percentage error occurred in the measurement of physical quantities A, B, C and D are 1%, 2%, 3% and 4% respectively. Then the maximum percentage of error in the measurement X, where $X = \frac{A^2 B^{1/2}}{C^{1/3} D^3}$, will be

(1) -10%

(2) 10%

(3) $\left(\frac{3}{13}\right)\%$

(4) 16%

92. Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1. On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2. The time taken by her to walk up on the moving escalator will be:

(1) $t_1 - t_2$

(2) $\frac{t_1 + t_2}{2}$

(3) $\frac{t_1 t_2}{t_2 - t_1}$

(4) $\frac{t_1 t_2}{t_2 + t_1}$

93. A car of mass 1000 kg moves in circular path with a constant speed of 16 m/s. It is turned by 90º after travelling 628 m on the road. The centripetal force acting on the car is

(1) 160 N

(2) 320 N

(3) 640 N

(4) 1280 N

94. One end of string of length *l* is connected to a particle of mass 'm' and the other end is connected to a small peg on a smooth horizontal table. If the particle moves in circle with speed 'v' the net force on the particle (directed towards centre) will be (T represents the tension in the string)

(1) Zero

(2) T

(3) $T + \frac{m\upsilon^2}{l}$

(4) $T - \frac{m\upsilon^2}{l}$

95. The upper half of an inclined plane of inclination θ is perfectly smooth while lower half is rough. A block starting from rest at the top of the plane will again come to rest at the bottom, if the coefficient of friction between the block and lower half of the plane is given by

(1) $\mu = 2\tan\theta$

(2) $\mu = \tan\theta$

(3) $\mu = \frac{1}{\tan\theta}$

(4) $\mu = \frac{2}{\tan\theta}$

96. Consider a drop of rain water having mass 1 g falling from a height of 1 km. It hits the ground with a speed of 50 m/s. Take 'g' constant with a value $10\,m/s^2$. The work by the

i) Gravitational force and the

ii) Resistive force of air is:

(1) (i) 10 J (ii) – 8.75 J

(2) (i) -10 J (ii) – 8.25 J

(3) (i) 1.25 J (ii) -8.25 J

(4) (i) 100 J (ii) 8.75 J

97. A rope is wound around a hollow cylinder of mass 3 kg and radius 40cm. What is the angular acceleration of the cylinder if the rope is pulled with a force of 30 N?

(1) $5\,m/s^2$

(2) $25\,m/s^2$

(3) $0.25\,rad/s^2$

(4) $25\,rad/s^2$

98. Two discs of same moment of inertia rotating about their regular axis passing through – centre and perpendicular to the plane of disc with angular velocities ω_1 and ω_2. They are brought into contact face to face coinciding the axis of rotation. The expression for loss of energy during this process is :

(1) $\frac{I}{8}(\omega_1-\omega_2)^2$

(2) $\frac{I}{2}(\omega_1+\omega_2)^2$

(3) $\frac{I}{4}(\omega_1-\omega_2)^2$

(4) $I(\omega_1-\omega_2)^2$

99. The distance covered by a particle undergoing SHM in one time period is (amplitude = A)

(1) Zero

(2) A

(3) 2A

(4) 4A

100. The acceleration due to gravity at a height 1 km above the earth is the same as at a depth of below the surface of earth. Then

(1) $d=2km$

(2) $d=\frac{1}{2}km$

(3) $d=1km$

(4) $d=\frac{3}{2}km$

101. When a block of mass M is suspended by a long wire of length L, the length of the wire become (L + *l*). The elastic potential energy stored in the extended wire is

(1) Mgl

(2) MgL

(3) $\frac{1}{2}Mgl$

(4) $\frac{1}{2}MgL$

102. A small sphere of radius 'r' falls from rest in a viscous liquid. As a result, heat is produced due to viscous force. The rate of production of heat when the sphere attains its terminal velocity, is proportional to

(1) r^3

(2) r^2

(3) r^4

(4) r^5

103. If 'M' is the mass of water that rises in a capillary tube of radius 'r', then mass of water which will rise in a capillary tube of radius '2r' is

(1) 4M

(2) M

(3) 2M

(4) $\frac{M}{2}$

104. If 'α' of a material is $10\times10^{-6}/^oC$ ratio of its densities at 10ºC and 30ºC is

(1) 1.06

(2) 1.006

(3) 1.0006

(4) 0.9998

105. During an experiment, an ideal gas is found to obey an additional law VP^2 = constant. The gas is initially at a temperature T and volume V. When it expands to 2V, the temperature becomes

(1) 2T

(2) $\sqrt{2}T$

(3) $\frac{T}{2}$

(4) $\frac{T}{\sqrt{2}}$

106. A piece of ice falls from a height h so that it melts completely. Only one-quarter of the heat produced is absorbed by the ice and all energy of ice gets converted into heat during its fall. The value of h is : [Latent heat of ice is $3.4 \times 10^5 J / Kg$]

(1) 34 km

(2) 544 km

(3) 136 km

(4) 68 km

107. A spherical black body with a radius of 12 cm radiates 450 watt power at 500 K. If the radius were halved and the temperature doubled, the power radiated in watt would be

(1) 1800

(2) 225

(3) 450

(4) 1000

108. A Carnot engine having an efficiency of $\frac{1}{10}$ as heat engine, is used as a refrigerator. If the work done on the system is 10J, the amount of energy absorbed from the reservoir at lower temperature is :

(1) 100 J

(2) 1 J

(3) 90 J

(4) 99J

109. Certain quantity of heat is supplied to a monatomic ideal gas which expands at constant pressure. The percentage of heat that is used to do work by the gas is

(1) 80%

(2) 20%

(3) 40%

(4) 60%

110. The *r.m.s.* speed of the molecules of a gas in a vessel is $400 ms^{-1}$. If half of the gas leaks out, at constant temperature, the *r.m.s.* speed of the remaining molecules will be

(1) 800 ms^{-1}

(2) $400\sqrt{2}\ ms^{-1}$

(3) 400 ms^{-1}

(4) 200 ms^{-1}

111. Mean kinetic energy per degree of freedom of gas molecules is

(1) $\frac{3}{2}KT$

(2) KT

(3) $\frac{1}{2}KT$

(4) $\frac{3}{2}RT$

112. Speed of a wave is 250 m/s. Frequency is 500 cycles/sec. The distance between the two points having phase difference $\pi / 3$ is

(1) 1m

(2) $\frac{1}{2}m$

(3) $\frac{1}{6}m$

(4) $\frac{1}{12}m$

113. The two nearest harmonics of a tube closed at one end and open at other end are 220 Hz and 260 Hz. What is the fundamental frequency of the system

(1) 40 Hz

(2) 10 Hz

(3) 20 Hz

(4) 30 Hz

114. An equiconvex lens has power P. It is cut into two symmetrical halves by a plane containing the principal axis. The power of one part will be

(1) 0

(2) $\frac{P}{2}$

(3) $\frac{P}{4}$

(4) P

115. In total internal reflection when the angle of incidence is equal to the critical angle for the pair of media in contact, what will be angle of refraction?

(1) Equal to angle of incidence

(2) 90º

(3) 180º

(4) 0º

116. Which colour of the light has the longest wavelength ?

(1) Green

(2) Violet

(3) Red

(4) Blue

117. The ratio of resolving powers of an optical microscope for two wave lengths $\lambda_1 = 4000\text{Å}$ and $\lambda_2 = 6000\text{Å}$ is

(1) 16 : 81

(2) 8 : 27

(3) 9 : 4

(4) 3 : 2

118. Two identical spheres of charges 8 mc and – 12 mc are at certain separation in between them in air then the force of attraction is 'F'. They are put in contact and again separated by the same distance in a medium of dielectric constant 2, then the force between them is .....

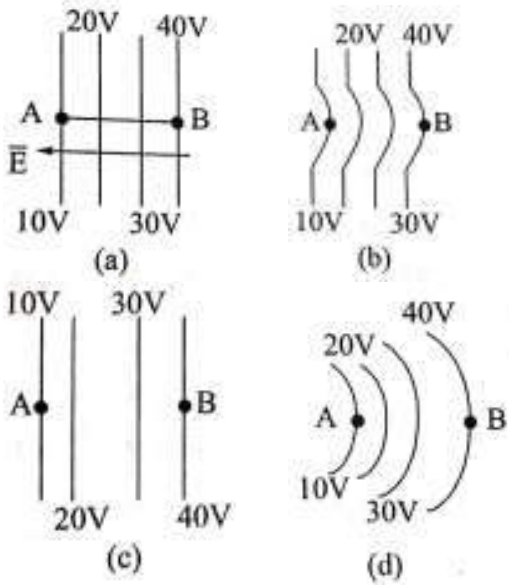
(1) $\frac{F}{48}$ attraction

(2) $\frac{F}{48}$ repulsion

(3) $\frac{3F}{4}$ repulsion

(4) $\frac{4F}{3}$ repulsion

119. The diagrams below show regions of equipotentials



A positive charge is moved from A to B in each diagram

(1) Maximum work is required to move q in figure (b)

(2) Maximum work is required to move q in figure (c)

(3) In all the four cases the work done is the same.

(4) Minimum work is required to move q in figure (a)

120. A capacitor is charged by a battery. The battery is removed and another identical uncharged capacitor is connected in parallel. The total electrostatic energy of resulting system

(1) Increases by a factor of 2

(2) Increases by a factor of 4

(3) Decreases by a factor of 2

(4) Remains the same

121. A cube has a resistance R between any two opposite faces. If the cube is split into 27 identical cubes, then the resistance between opposite faces of small cube.

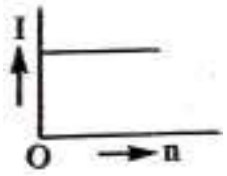
(1) 3R

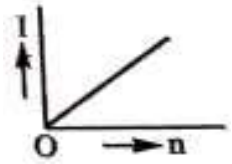
(2) R/3

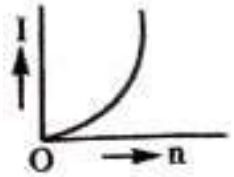
(3) R/27

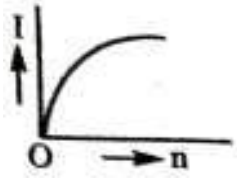
(4) 27 R

122. A battery consists of a variable number 'n' of identical cells (having internal resistance 'r' each) which are connected in series. The terminals of the battery are short –circuited and the current I is measured. Which of the graphs shows the correct relationship between I and n?

(1) 

(2) 

(3) 

(4) 

123. A circular coil ABCD carrying a current is placed in a uniform magnetic field. If the magnetic force on the segment AB is $\vec{F}$, the force on the remaining segment BCDA is

(1) $\vec{F}$

(2) $-\vec{F}$

(3) $3\vec{F}$

(4) $-3\vec{F}$

124. In a galvanometer, 5% of the total current in the circuit passes through it. If the resistance of the galvanometer is G, the resistance of the shunted galvanometer is

(1) 19 G

(2) $\frac{G}{19}$

(3) 20 G

(4) $\frac{G}{20}$

125. The magnetic susceptibility is negative for

(1) Paramagnetic material only

(2) Ferromagnetic material only

(3) Paramagnetic and ferromagnetic materials

(4) Diamagnetic material only

126. At a certain place, the horizontal component of earth's magnetic field is $\frac{1}{\sqrt{3}}$ times the vertical component. The angle of dip at that place is

(1) 30º

(2) 45º

(3) 60º

(4) 90º

127. The coefficient of mutual inductance, when magnetic flux changes by $2\times10^{-2}Wb$ and current changes by 0.01A is

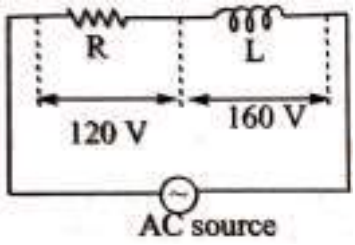
(1) 2H

(2) 4H

(3) 3H

(4) 8H

128. The circuit given in figure has a resistanceless chock coil L and resistance R. The voltage across R and L are given in figure. The virtual value of the applied voltage is



(1) 100 V

(2) 200 V

(3) 300 V

(4) 400 V

129. The amplitudes of electric and magnetic fields in an electromagnetic wave are related to each other as

(1) $E_0 = B_0$

(2) $E_0 = cB_0$

(3) $E_0 = B_0 / c$

(4) $E_0 = 2B$

130. When the light of frequency $2v_0$ (where v_0 is threshold frequency), is incident on a metal plate, the maximum velocity of electrons emitted is v_1. When the frequency of the incident radiation is increased to $5v_0$, the maximum velocity of electrons emitted from the same plate is v_2. The ratio of v_1 to v_2 is

(1) 1 : 2

(2) 1 : 4

(3) 2 : 1

(4) 4 : 1

131. An electron is accelerated through a potential difference of 10,000 V. Its de Broglie wavelength is (nearly) $\left(m_c = 9\times10^{-31}kg\right)$

(1) $12.2\times10^{-14}m$

(2) $12.2nm$

(3) $12.2\times10^{-13}m$

(4) $12.2\times10^{-12}m$

132. The total energy of an electron in an atom in an orbit as – 3.4 eV. Its kinetic and potential energies are respectively

(1) 3.4eV, -6.8 eV

(2) 3.4 eV, 3.4 eV

(3) -3.4 eV, -3.4 eV

(4) -3.4 eV, -6.8 eV

133. α - particle consists of

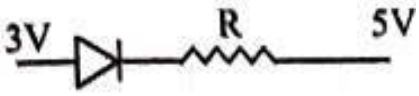
(1) 2 electrons and 4 protons only

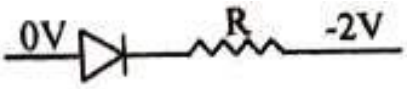
(2) 2 protons only

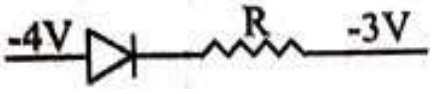
(3) 2 protons and 2 neutrons only

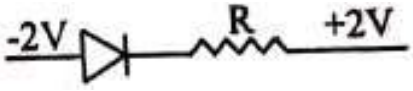
(4) 2 electrons, 2 protons and 2 neutrons

134. Which one of the following represents forward bias diode?

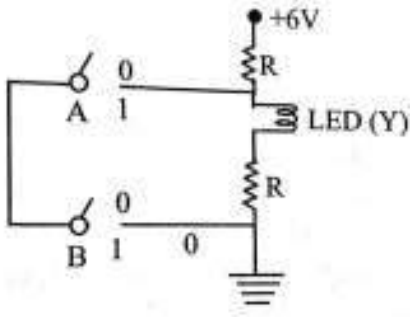
(1) 

(2) 

(3) 

(4) 

135. The correct Boolean operation represented by the circuit diagram drawn is



(1) NAND

(2) NOR

(3) AND

(4) OR

136. A solution of urea boils at 100.26^0C at the atmospheric pressure. If K_b and K_f for water are 0.52 K.kg.mol^{-1} and 1.86 K.kg.mol^{-1} respectively. Then the above solution will freeze at:

(1) -1.86^0C

(2) -0.465^0C

(3) -0.93^0C

(4) -0.52^0C

137. The atomic weights of two elements A and B are 40 and 80 respectively. If x g of A contains y atoms, how many atoms are present in 2x g of B?

(1) $\frac{y}{2}$

(2) y

(3) $\frac{y}{4}$

(4) 2y

138. The wave length of the shortest wavelength transition in Balmer series of atomic hydrogen will be:

(1) $4215\,\text{Å}$

(2) $1437\,\text{Å}$

(3) $3942\,\text{Å}$

(4) $3647\,\text{Å}$

139. The ratio of $(E_2 - E_1)$ to $(E_4 - E_3)$ for the hydrogen atom is approximately equal to

(1) 10

(2) 15

(3) 17

(4) 12

140. At very high pressures, the compressibility factor of one mole of a gas is given by:

(1) $\frac{Pb}{RT}$

(2) $1+\frac{Pb}{RT}$

(3) $1-\frac{Pb}{RT}$

(4) $1-\frac{b}{(VRT)}$

141. For a particular reaction, $\Delta H^o = -40\,\text{kJ}$ and $\Delta S^o = -200\,\text{J K}^{-}\text{mol}^{-1}$. The reaction is spontaneous at:

(1) 200K

(2) at all temperatures

(3) below 200K

(4) above 200K

142. Vapour density of the equilibrium mixture of the reaction $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$ is 50. The extent of dissociation of $SO_2Cl_2(g)$ is

(1) 33.33 %

(2) 35.0 %

(3) 30.0 %

(4) 66.67 %

143. pH of NH_4Cl(aq) is:

(1) 7

(2) >7

(3) 14

(4) <7

144. K_{sp} of $SrF_2(s)$ in water is 3.2×10^{-11}. Then the solubility of $SrF_2(s)$ in water is:

(1) $3.2 \times 10^{-9}(M)$

(2) $2 \times 10^{-4}(M)$

(3) $4 \times 10^{-4}(M)$

(4) $3 \times 10^{-4}(M)$

145. In a normal spinal types structure, the oxide ions are arranged in ccp, where as 1/8 tetrahedral holes are occupied by Zn^{2+} ions and 50% of octahedral holes are occupied by Fe^{+3} ions. The formula of the compound is

(1) $Zn_2Fe_2O_4$

(2) $ZnFe_2O_3$

(3) $ZnFe_2O_4$

(4) $ZnFe_2O_2$

146. The ratio of the value of any colligative property for $K_4[Fe(CN)_6]$ to that of $Fe_4[Fe(CN)_6]_3$ solution is nearly

(1) 0.62

(2) 1.4

(3) 0.71

(4) 1.2

147. The reaction

$5H_2O_5 + XClO_2 + 2OH^- \rightarrow XCl^- + YO_2 + 6H_2O$

is balanced, if

(1) X = 5, Y = 2

(2) X = 2, Y = 5

(3) X = 4, Y = 10

(4) X = 5, Y = 5

148. The molar conductances of NaCl, HCl and CH_3COONa at infinite dilution are 126.45, 426.16 and 91 $ohm^{-1} cm^2 mol^{-1}$ respectively. The molar conductance of CH_3COOH at infinite dilution is

(1) 201.28 $ohm^{-1} cm^2 mol^{-1}$

(2) 698.28 $ohm^{-1} cm^2 mol^{-1}$

(3) 390.72 $ohm^{-1} cm^2 mol^{-1}$

(4) 540.48 $ohm^{-1} cm^2 mol^{-1}$

149. The standard reduction potential for Fe^{2+}/Fe and Sn^{2+}/Sn electrodes are -0.44 and – 0.14 volt respectively. The standard EMF of cell is

(1) – 0.39 V

(2) – 0.58 V

(3) + 0.58 V

(4) + 0.30 V

150. Sucrose is not reducing sugar since

(1) It is chemically stable

(2) It is built up of a fructose unit

(3) It contains no free $-CHO$ or $>C=O$ group adjacent to a $>CHOH$ group

(4) It is optically active

151. The gaseous reaction $A(g) \rightarrow 2B(g) + C(g)$ is found to be first – order with respect to A. If the reaction is started with $p_A = 90\,\text{torr}$ the pressure after 10 min is found to be 180 torr. The rate constant of the reaction is

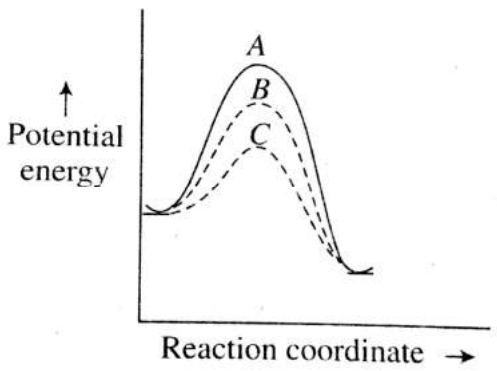
(1) $0.0693\,\text{min}^{-1}$

(2) $2.30\,\text{min}^{-1}$

(3) $0.0345\,\text{min}^{-1}$

(4) $4.60 \times 10^{-3}\,\text{min}^{-1}$

152. In homogeneous catalytic reactions, there are three alternative paths A, B and C (shown in the figure). Which one of the following indicates the relative ease with which the reaction can take place ?



(1) A > B > C

(2) C > B > A

(3) B > C > A

(4) A = B = C

153. The gold number of A, B, C and D are 0.04, 0.002, 10 and 25 respectively. Protective power of A, B, C and D are in order

(1) A > B > C > D

(2) B > A > C > D

(3) D > C > B > A

(4) C > A > B > D

154. The following equation represents a method of purification of nickel by

$$\underset{\text{Im pure}}{\text{Ni}} + 2\text{CO} \xrightarrow{320\,\text{K}} \text{Ni(CO)}_4 \xrightarrow{420\,\text{K}} \underset{\text{Pure}}{\text{Ni}} + 4\text{CO}$$

the method is

(1) Cupellation

(2) Mond's process

(3) Electrical furnace

(4) Muffle furnace

155. Which one of the following is a chain growth polymer?

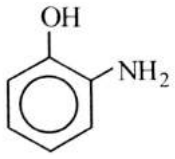
(1) Dacron

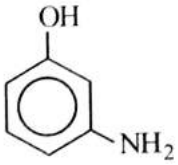
(2) Nylon – 6, 6

(3) Nylon – 6

(4) Polypropylene

156. Which of the following gives paracetamol on acetylation?

(1) 

(2) 

(3) 

(4) 

157. The brown ring test for NO_2^- and NO_3^- is due to the formation of complex ion with formula

(1) $\left[Fe(H_2O)_6\right]^{2+}$

(2) $\left[Fe(NO)(CN)_5\right]^{2+}$

(3) $\left[Fe(H_2O)_5NO\right]^{2+}$

(4) $\left[Fe(H_2O)(NO)_5\right]^{2+}$

158. Correct match is

(1) $HClO_4 > HClO_3 > HClO_2 > HClO$ – acidic nature

(2) $Cl_2 > Br_2 > F_2 > I_2$ – Bond dissociation enthalpy

(3) $HClO > HClO_2 > HClO_3 > HClO_4$ – Oxidising nature

(4) All are correct

159. Which amongst the following is the strongest acid?

(1) $CHBr_3$

(2) $CHCl_3$

(3) $CH(CN)_3$

(4) CHI_3

160. Which of the following acids has the smallest dissociation constant

(1) $CH_3CHFCOOH$

(2) FCH_2CH_2COOH

(3) $BrCH_2CH_2COOH$

(4) $CH_3CHBrCOOH$

161. The total number of isomers for a square planar complex [M(F)(C*l*)(SCN)(NO_2)] is

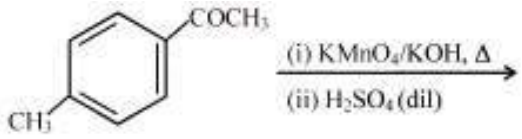
(1) 6

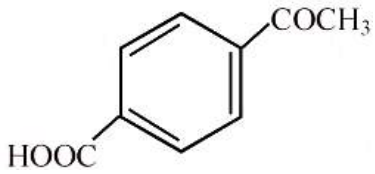
(2) 8

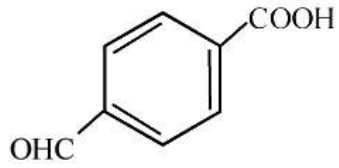
(3) 12

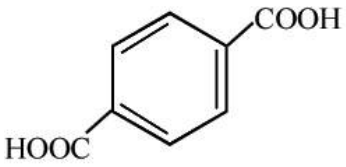
(4) 4

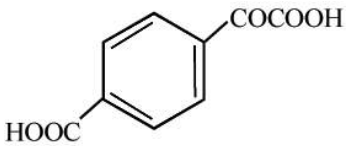
162. The major product of the following reaction is



(1) 

(2) 

(3) 

(4) 

163. n-propyl cyanide and Isopropyl cyanide are

(1) Chain isomers

(2) Positional isomers

(3) Functional Isomers

(4) Metamers

164. Which of the following is used as arsenic purifier in the contact process of manufacture of H_2SO_4?

(1) $Na_2CO_3.10H_2O$

(2) $FeSO_4.7H_2O$

(3) $Fe_2O_3.xH_2O$

(4) $Al_2O_3.2H_2O$

165. Match the following items in column I with the corresponding items in column II.

Column I	Column II
(i) $Na_2CO_3.10H_2O$	(A) Portland cement ingredient
(ii)$Mg(HCO_3)_2$	(B) Castner-Kellner process
(iii) NaOH	(C) Solvay process
(iv) $Ca_3Al_2O_6$	(D)Temporary hardness

(1) i $\rightarrow$ D; ii $\rightarrow$ A; ii $\rightarrow$ B; iv $\rightarrow$ C

(2) i $\rightarrow$ B; ii $\rightarrow$ C; iii $\rightarrow$ A; iv $\rightarrow$ D

(3) i $\rightarrow$ C; ii $\rightarrow$ B; iii $\rightarrow$ D; iv $\rightarrow$ A

(4) i $\rightarrow$ C; ii $\rightarrow$ D; iii $\rightarrow$ B; iv $\rightarrow$ A

166. The correct order for acid strength of compounds: CH $\equiv$ CH, CH_3–C $\equiv$ CH and $CH_2 = CH_2$ is as follows

(1) HC $\equiv$ CH >CH_2= CH_2 > CH_3 – C$\equiv$ CH

(2) CH_3 – C $\equiv$ CH > CH_2 = CH_2>HC$\equiv$ CH

(3) HC $\equiv$ CH > CH_3 – C $\equiv$ CH >CH_2 = CH_2

(4) CH_3 – C $\equiv$ CH > HC $\equiv$ CH > CH_2=CH_2

167. $Mn_2(CO)_{10}$ is an organometallic compound due to the presence of

(1) C–O bond

(2) Mn – Mn bond

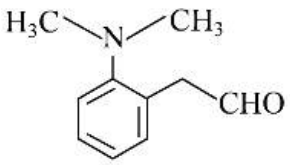
(3) Mn – O bond

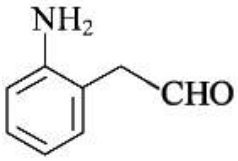
(4) Mn – C bond

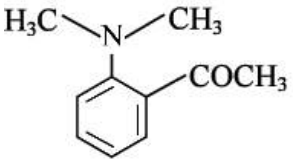
168. The tests performed on compound X and their inferences are:

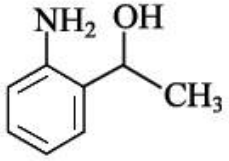
	Test	Inference
(a)	2, 4-DNP test	Coloured Precipitate
(b)	Iodoform test	Yellow Precipitate

Compound 'X' is –

(1) 

(2) 

(3) 

(4) 

169. In the dichromate anion

(1) 4 Cr - O - bonds are equivalent

(2) 6 Cr - O bonds are equivalent

(3) All Cr - O bonds are equivalent

(4) All Cr - O - bonds are non-equivalent

170. In borax, the number of B–O–B links and B–OH bonds present respectively are

(1) Five and four

(2) Four and five

(3) Three and four

(4) Five and five

171. Consider the reactions

I. $Zn + conc.HNO_3(hot) \rightarrow Zn(NO_3)_2 + H_2O + X$

II. $Zn + dil.HNO_3(cold) \rightarrow Zn(NO_3)_2 + H_2O + Y$

compounds X and Y are respectively

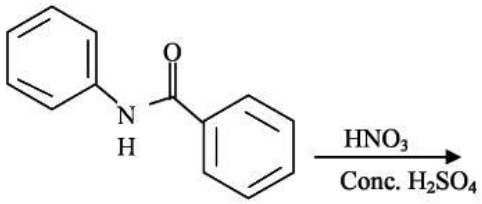
(1) $N_2O,\ NO$

(2) NO_2, N_2O

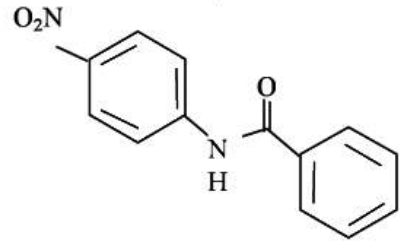
(3) $N_2,\ N_2O$

(4) $NO_2,\ NO$

172. What will be the major product in the following mononitration reaction?



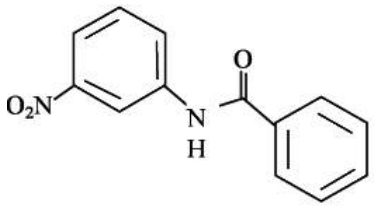
(1)



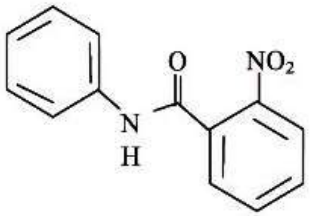
(2)



(3)



(4)



173. Which one of the following is used to block the ends of polymeric silicone chain

(1) R_3SiCl

(2) $RSiCl_3$

(3) R_2SiCl_2

(4) R_4Si

174. The magnetic moment of an octahedral homoleptic Mn(II) complex is 5.9 BM. The suitable ligand for this complex is

(1) CN^-

(2) ethylenediamine

(3) Cl^-

(4) CO

175. Which of the following can reduce a carboxylic acid to primary alcohol

(1) $NaBH_4$

(2) $LiAlH_4$

(3) DiBAL-H

(4) HI / Red P

176. Incorrect among the following ?

(1) $XeO_3 - sp^3 - 1l.P - Pyramidal$

(2) $XeF_2 - sp^3d - 3l.P - Linear$

(3) $XeF_6 - sp^3d^3 - 1l.P - Distorted\ Octahedral$

(4) $XeF_4 - sp^3d - 2l.P - Square\ planar$

177. Atomic number of an element in long form periodic table is 114. Then its group number and period number respectively are

(1) 17th group, 7th period

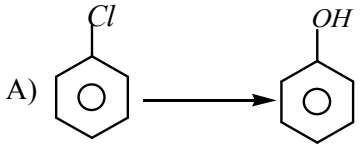
(2) 14th group, 7th period

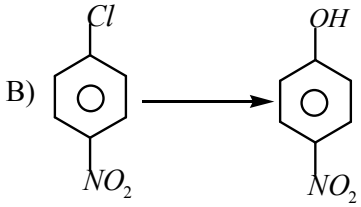
(3) 15th group, 7th period

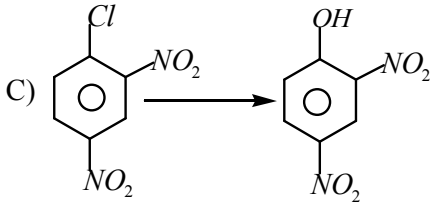
(4) 16th group, 7th period

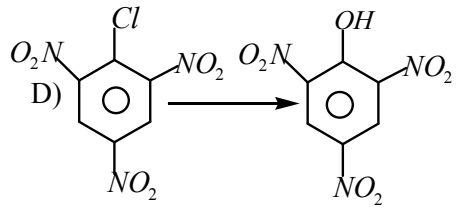
178. Match the following conditions of the reactions given in column I with Column II and choose the correct option from the codes given below

Column I

A) 

B) 

C) 

D) 

Column II

1) $NaOH, 443K, H^{\oplus}$

2) $NaOH, 368K, H^{\oplus}$

3) Warm H_2O

4) NaOH, $623K, 300atm$, $H^{\oplus}$

Codes:

(1) A-3, B-1, C-2, D-4

(2) A-1, B-3, C-2, D-4

(3) A-4, B-1, C-2, D-3

(4) A-4, B-1, C-3, D-2

179. Statement A: Double bond in C_2 molecule consists of both Pi bonds.

Statement B: C_2 molecule is paramagnetic

(1) Both statement A and statement B are incorrect

(2) Both statement A and statement B are correct

(3) Statement A is correct but statement B is incorrect

(4) Statement A is incorrect but statement B is correct

180. Incorrect match among the following is

(1) $CO_{(g)} + H_2O_{(g)} \xrightarrow[673k]{FeCrO_4} CO_{2(g)} + H_{2(g)}$ - Water gas shift equation

(2) $C_{(s)} + H_2O_{(g)} \xrightarrow{1270K} CO_{(g)} + H_{2(g)}$ - Coal gasification

(3) $Na_6P_6O_{18}$ - Permutit

(4) $R-CH=CH_2 + CO + H_2 \rightarrow RCH_2CH_2CHO$ - Hydroformylation

SRI CHAITANYA EDUCATIONAL INSTITUTIONS,INDIA

A.P,TELANGANA,KARNATAKA,TAMILNADU,MAHARASHTRA,DELHI,RANCHI,CHANDIGARH

SR ELITE, AIIMS S60 & NEET MPL NEET GRAND TEST – 1 KEY Date : 10-01-2020

BOT : 1, 2, 5, 7, 9, 11, 13, 14, 15, 19, 21, 23, 25, 27, 29, 30, 31, 35, 36, 39, 41, 43, 44, 47, 49, 51, 53, 55, 57, 59, 61, 62, 63, 67, 68, 71, 72, 75, 77, 79, 80, 83, 85, 87, 89

ZOO : 3, 4, 6, 8, 10, 12, 16, 17, 18, 20, 22, 24, 26, 28, 32, 33, 34, 37, 38, 40, 42, 45, 46, 48, 50, 52, 54, 56, 58, 60, 64, 65, 66, 69, 70, 73, 74, 76, 78, 81, 82, 84, 86, 88, 90

BIOLOGY

1) **2**	2) **4**	3) **3**	4) **4**	5) **3**	6) **2**	7) **3**	8) **3**	9) **3**	10) **4**
11) **1**	12) **1**	13) **4**	14) **2**	15) **4**	16) **1**	17) **1**	18) **1**	19) **3**	20) **2**
21) **1**	22) **3**	23) **3**	24) **1**	25) **4**	26) **3**	27) **3**	28) **2**	29) **2**	30) **3**
31) **3**	32) **4**	33) **4**	34) **4**	35) **2**	36) **3**	37) **2**	38) **3**	39) **1**	40) **1**
41) **2**	42) **3**	43) **3**	44) **2**	45) **4**	46) **1**	47) **4**	48) **4**	49) **3**	50) **2**
51) **4**	52) **1**	53) **1**	54) **1**	55) **4**	56) **4**	57) **3**	58) **4**	59) **2**	60) **3**
61) **4**	62) **4**	63) **3**	64) **4**	65) **3**	66) **2**	67) **2**	68) **2**	69) **4**	70) **3**
71) **2**	72) **4**	73) **1**	74) **1**	75) **1**	76) **2**	77) **2**	78) **1**	79) **3**	80) **2**
81) **2**	82) **2**	83) **2**	84) **3**	85) **3**	86) **2**	87) **2**	88) **3**	89) **4**	90) **4**

PHYSICS

91) **4**	92) **4**	93) **3**	94) **2**	95) **1**	96) **1**	97) **4**	98) **3**	99) **4**	100) **1**
101) **3**	102) **4**	103) **3**	104) **3**	105) **2**	106) **3**	107) **1**	108) **3**	109) **3**	110) **3**
111) **3**	112) **4**	113) **3**	114) **4**	115) **2**	116) **3**	117) **4**	118) **2**	119) **3**	120) **3**
121) **1**	122) **1**	123) **2**	124) **4**	125) **4**	126) **3**	127) **1**	128) **2**	129) **2**	130) **1**
131) **4**	132) **1**	133) **3**	134) **2**	135) **1**					

CHEMISTRY

136) **3**	137) **2**	138) **4**	139) **2**	140) **2**	141) **3**	142) **2**	143) **4**	144) **2**	145) **3**
146) **3**	147) **2**	148) **3**	149) **4**	150) **3**	151) **1**	152) **2**	153) **2**	154) **2**	155) **4**
156) **3**	157) **3**	158) **4**	159) **3**	160) **3**	161) **3**	162) **3**	163) **1**	164) **3**	165) **4**
166) **3**	167) **4**	168) **3**	169) **2**	170) **1**	171) **2**	172) **1**	173) **1**	174) **3**	175) **2**
176) **4**	177) **2**	178) **3**	179) **3**	180) **3**					